

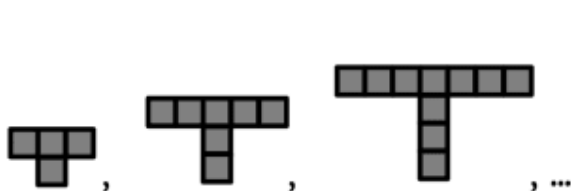


Finding the n th term

- 1 The n^{th} term is the position of a term in a sequence (n is the term number). It can be used to calculate any term so is also known as a _____ for finding the n^{th} term. Tick **1** correct answer

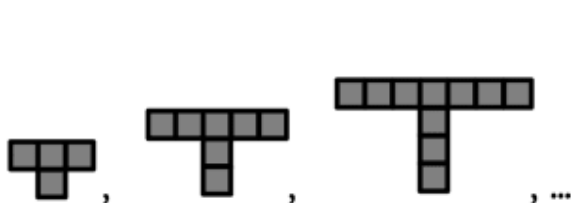
- ☒ formula
☐ sequence
☐ variable

- 2 What is the term-to-term rule for this pattern sequence? Tick **1** correct answer



- ☐ Multiply by 3
☐ Goes up by 1
☐ Multiply by 3 and add 1
☒ Add 3

- 3 What is the position-to-term rule for this pattern sequence? Tick **1** correct answer



- ☐ Multiply the term number by 3
☐ Add 1 to the previous term
☒ Multiply the term number by 3 and add 1
☐ Add 3 to the previous term



4 Which of these are arithmetic (linear) sequences? Tick **3** correct answers

☐ 1, 2, 4, 8, ...

☐ 1, 4, 8, 13, ...

☒ 1, 4, 7, 10, ...

☒ 10, 7, 4, 1, ...

☒ 0.4, 0.7, 1, 1.3, ...

5 This pattern represents people seated around an increasing number of tables. If you were asked to find the number of people around 50 tables, which calculation would you do? Tick **1** correct answer



☐ 50×4

☐ $50 \times 2 + 4$

☐ 50×6

☐ $4 + 4 + 4 + 4 + 4 + \dots$ fifty times.

☒ $50 \times 4 + 2$

6 Order these arithmetic sequences in terms of the size of their common difference. Start with the greatest common difference. Use numbers to show the correct order

2	124, 133, 142, 151, ...
6	1896, 1888, 1880, 1872, ...
5	1852, 1859, 1866, 1873, ...
4	-5, 2.5, 10, 17.5 ...
1	-101, -87, -73, -59, ...
7	2190, 2180, 2170, 2160, ...
3	-8, 0, 8, 16, 24, ...

